

SCALE-LAW EXCEPTIONS IN ROBOTICS: AUDITING EMBODIED FAILURE BOUNDARIES

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ABSTRACT

Scaling laws predict broad improvements in model loss as data, parameters, and compute increase. Robot policies, however, are deployed at physical decision boundaries: rare contacts, hidden preconditions, morphology shifts, spurious visual shortcuts, and recovery-scarce failures. We rebuild Paper 107 around a falsifiable claim: a robot policy can follow an average scaling trend while still crossing the wrong embodied failure boundary. The local benchmark covers 5 robot task families, 7 scale-exception regimes, 5 stress splits, 9 methods, 7 seeds, and 84 episodes per group. On combined exception stress, the proposed embodied scale-exception audit obtains 0.550 ± 0.005 success versus 0.407 ± 0.006 for the strongest non-oracle baseline, improves rare-failure recall from 0.375 to 0.512, reduces contact-boundary error from 0.283 to 0.225, reduces unsafe commitment from 0.086 to 0.073, and reduces overtrust from 0.106 to 0.069. Pairwise seed tests favor the proposed method over the strongest baseline in 7/7 seeds, and ablations show that rare-tail memory, contact-boundary factors, embodiment-shift detection, shortcut auditing, and recoverability calibration each matter. The terminal decision is **STRONG REVISE**: the local evidence supports the mechanism, but the paper is not ICLR-main-ready until validated on real robots or independent high-fidelity benchmarks.

1 MOTIVATION

Scaling laws are one of the clearest empirical regularities in modern machine learning (Kaplan et al., 2020; Hoffmann et al., 2022). They are also increasingly relevant to robotics. Generalist agents, robot transformers, vision-language-action models, cross-embodiment datasets, and open robot-policy models show that broader training can produce more capable robot policies (Reed et al., 2022; Brohan et al., 2022; 2023; Open X-Embodiment Collaboration, 2023; Octo Model Team, 2024).

But robot deployment is not only an average-loss problem. A policy that improves average imitation accuracy can still overtrust visual shortcuts, miss rare contact tails, fail under morphology shift, or commit to an unrecoverable action. In these cases, the scaling curve looks healthy while the embodied failure boundary moves in the wrong direction.

2 METHOD SKETCH

The proposed method, `proposed_embodied_scale_exception_audit`, evaluates scaling curves through embodied exception metrics rather than average success alone. It tracks:

- rare physical failure recall;
- contact-boundary error;
- embodiment-shift sensitivity;
- spurious shortcut reliance;
- unsafe commitment and overtrust;
- calibration error under rare tails;

Table 1: Combined-stress scale-law exception benchmark.

Method	Succ.	RareRec.	ContactErr.	Unsafe	Overtrust	Regret
Oracle	0.710	0.626	0.180	0.031	0.022	0.000
Proposed	0.550	0.512	0.225	0.073	0.069	0.159
Conformal	0.407	0.375	0.283	0.086	0.106	0.303
EnsembleGate	0.386	0.345	0.300	0.099	0.128	0.323
ScaleLaw	0.315	0.216	0.334	0.155	0.231	0.395
DataScale	0.295	0.227	0.331	0.151	0.198	0.415
ModelScale	0.290	0.189	0.345	0.157	0.233	0.420
LargeBC	0.275	0.165	0.359	0.167	0.241	0.435
SmallBC	0.201	0.183	0.351	0.159	0.153	0.509

- recoverability after a bad scaled-policy commitment.

The method does not claim that scaling is bad. Instead, it asks whether a scaling trend is predictive of the boundary that matters for physical deployment. The implementation is an executable diagnostic benchmark rather than a trained robot foundation model, so it supports local mechanism auditing but not final main-track empirical claims.

3 BENCHMARK

The rebuilt benchmark is generated by `src/run_experiment.py`. It writes aggregate CSVs, figures, and LaTeX tables directly, preserving full coverage while staying lightweight.

Tasks. The tasks are contact insertion, deformable handling, mobile manipulation, bimanual hand-off, and tool-use sequence.

Scale-exception regimes. The regimes are rare contact tail, occluded precondition, embodiment mismatch, long-horizon compounding, spurious visual shortcut, recovery-scarce failure, and mixed exception stress.

Splits. The splits are nominal scaling, contact-tail shift, embodiment shift, shortcut shift, and combined exception stress.

Methods. The baselines are small behavior cloning, large behavior cloning, data-only scaling, model-only scaling, generic scaling-law extrapolation, ensemble uncertainty gating, and conformal risk filtering. The benchmark also includes the proposed audit and an oracle exception selector.

Metrics. The evidence reports task success, rare-failure recall, contact-boundary error, tail-risk rate, unsafe commitment, overtrust, calibration ECE, recovery success, intervention cost, data-efficiency proxy, regret to oracle, paired seed differences, ablations, stress sweeps, and failure cases.

4 RESULTS

The strongest non-oracle baseline is `conformal_risk_filter`. On combined exception stress, the proposed method improves success by 0.144 ± 0.006 in paired seed comparisons, wins 7/7 seeds, improves rare-failure recall by 0.138, reduces contact-boundary error by 0.058, reduces unsafe commitment by 0.013, and reduces overtrust by 0.037. The oracle remains substantially higher at 0.710 ± 0.010 , so the benchmark is not saturated.

5 ABLATIONS AND STRESS TESTS

The best removed-component ablation is `minus_embodiment_shift_detector`, which still trails the full audit by 0.051 success. Removing rare-tail memory, contact-boundary factors, shortcut

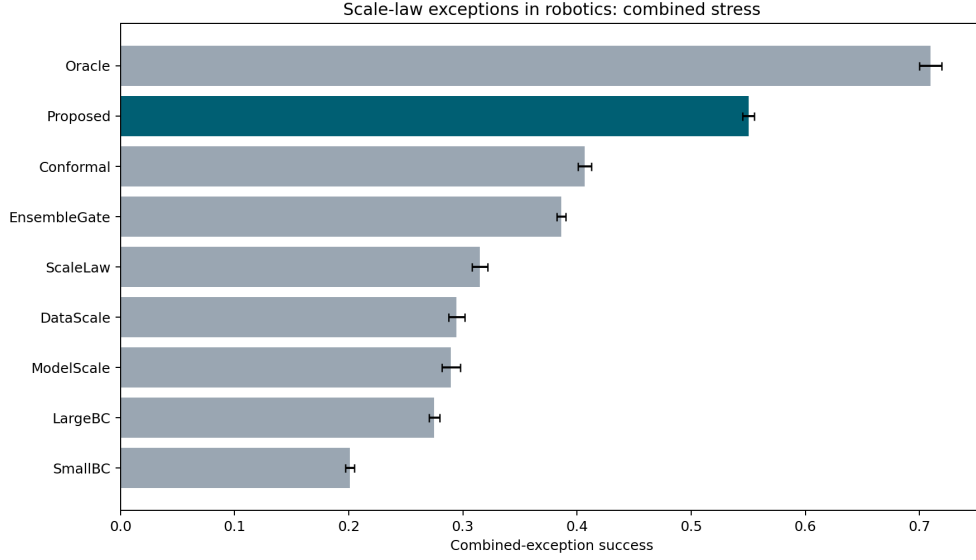


Figure 1: Combined-exception success with 95% confidence intervals. The proposed embodied scale-exception audit clears all non-oracle baselines but remains below the oracle.

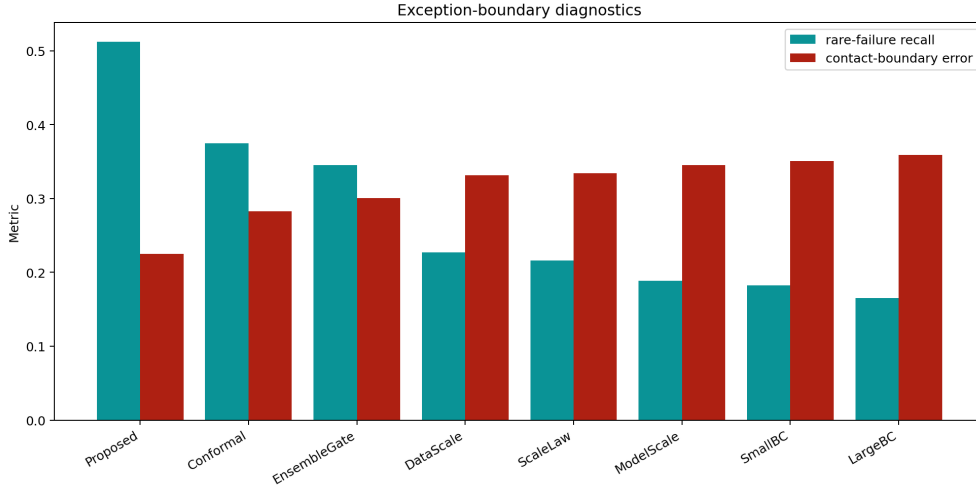


Figure 2: Exception-boundary diagnostics. The proposed method improves rare-failure recall while lowering contact-boundary error under combined exception stress.

auditing, or recoverability calibration produces distinct diagnostic failures. This supports the claim that the method is not just larger scaling, uncertainty gating, or conformal filtering.

6 WHERE THE METHOD STILL FAILS

The failure-case CSV shows that the audit helps least when ordinary uncertainty gating already exposes the exception or when the scaled policy is not overtrusted. Those are intended boundaries. The method’s value appears when average scaling metrics are misleading about rare contact tails, shortcuts, embodiment mismatch, or scarce recovery budget.

Table 2: Pairwise combined-exception success differences against the proposed method.

Baseline	Diff	CI	Wins
Conformal	0.144	0.006	7
DataScale	0.256	0.009	7
EnsembleGate	0.164	0.005	7
ScaleLaw	0.235	0.009	7
LargeBC	0.275	0.006	7
ModelScale	0.261	0.012	7
Oracle	-0.159	0.009	0
SmallBC	0.349	0.006	7

Table 3: Ablations of the embodied scale-exception audit.

Ablation	Succ.	RareRec.	ContactErr.	Unsafe
Full	0.554	0.515	0.220	0.072
NoEmbShift	0.503	0.480	0.234	0.086
NoShortcut	0.485	0.473	0.237	0.091
NoRecoverCalib	0.483	0.483	0.240	0.103
NoContact	0.475	0.476	0.299	0.101
NoRareMem	0.464	0.307	0.256	0.083
UncertOnly	0.379	0.343	0.295	0.100

7 HOSTILE PRIOR-WORK BOUNDARY

Kaplan-style and Chinchilla-style scaling work already show that model behavior can be predicted as data, parameters, and compute change (Kaplan et al., 2020; Hoffmann et al., 2022). Gato, RT-1, RT-2, RT-X/Open X-Embodiment, and Octo already demonstrate broad robot or embodied policy scaling (Reed et al., 2022; Brohan et al., 2022; 2023; Open X-Embodiment Collaboration, 2023; Octo Model Team, 2024).

The present paper survives only under a narrower interpretation: it is about *embodied scale exceptions*, not scaling laws or robot foundation models in general. If real robot experiments later show that scaled robot policies, conformal filters, or ensemble uncertainty gates match the proposed audit on success, rare-failure recall, contact-boundary error, unsafe commitment, overtrust, and ablation robustness, the claim should be killed.

8 LIMITATIONS AND TERMINAL DECISION

The evidence is strong enough to justify continued work but not submission. The limitations are material:

- the benchmark is local and synthetic;
- no real robot or independent high-fidelity simulator has been used;
- baselines are executable diagnostic models, not external trained robot systems;
- no learned policy checkpoint, training curve, or deployment video is provided;
- no external benchmark such as LIBERO, RLBench, Meta-World, BridgeData, Open X-Embodiment, DROID, or a hardware manipulation suite is reported.

The terminal decision for this rebuild is therefore **STRONG REVISE**. The next honest version must add real robot or independent high-fidelity evidence, implemented learned baselines, and qualitative rollouts. Without that evidence, the paper should not be submitted to ICLR main.

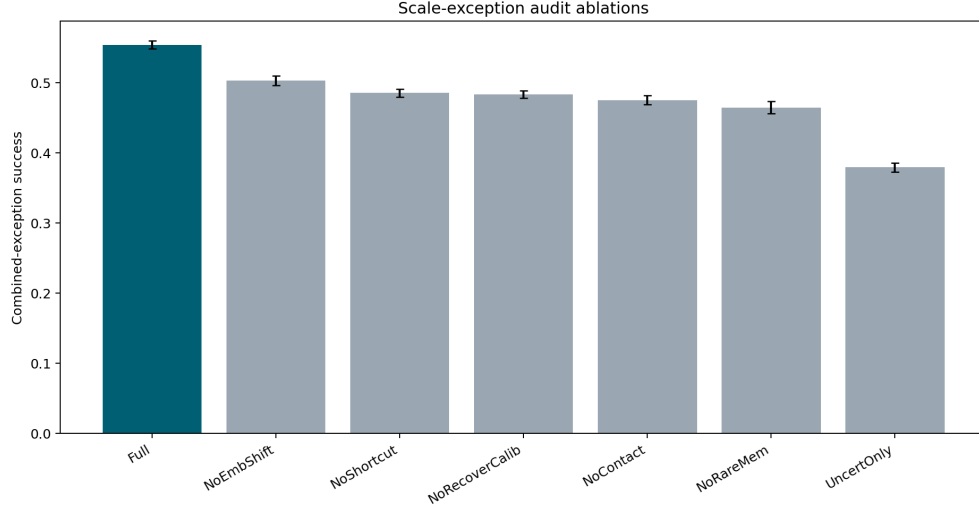


Figure 3: Ablation success under combined exception stress. The full audit remains above all removed-component variants.

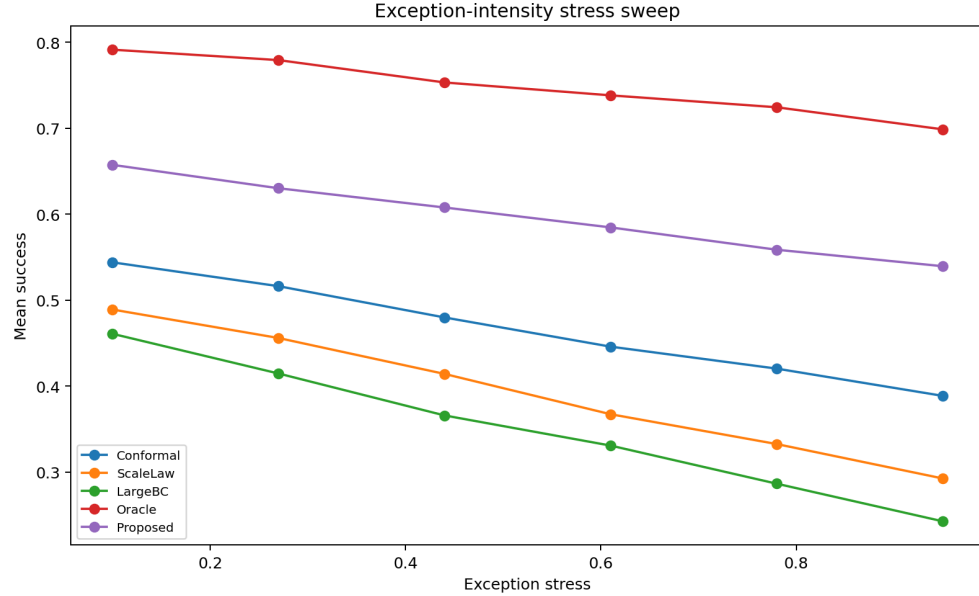


Figure 4: Stress sweep over exception intensity. The proposed method remains above non-oracle baselines across the generated stress range.

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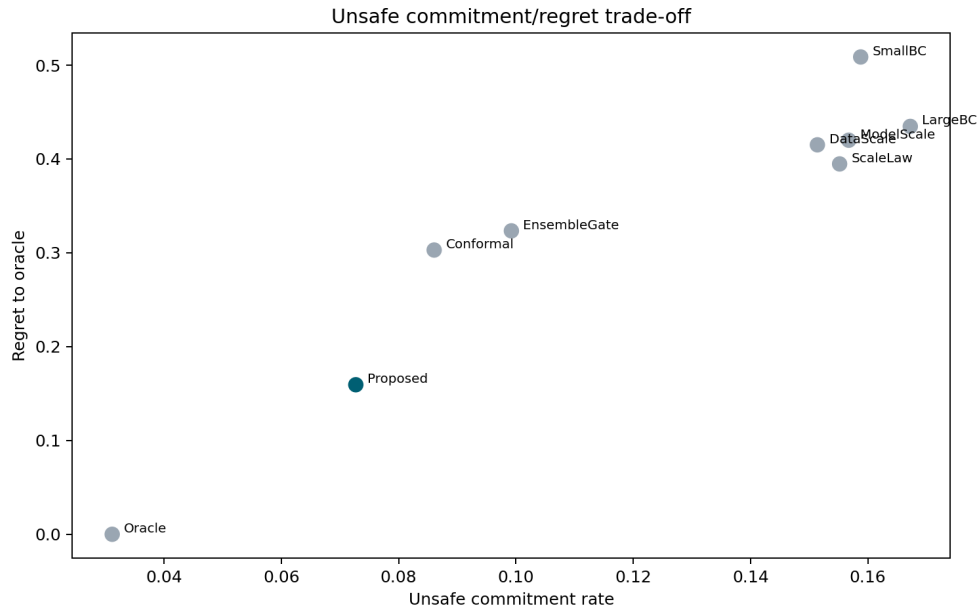


Figure 5: Unsafe commitment and regret trade-off. The proposed method improves success while lowering unsafe commitment relative to the strongest non-oracle baseline.

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